

SkillBridge Africa

A trust-first freelance marketplace connecting verified African university students with real, paid client work — with escrow-backed payments at its core.

Live skillbridge-africa-193.netlify.app

Tagline Your Skills. Our Bridge.

Functional MVP · Live

01 Executive summary

SkillBridge Africa is a marketplace where clients post tasks and verified students bid on and deliver them. An admin layer verifies students, oversees assignments, and safeguards payments held in escrow — so both sides can transact with confidence.

The product is built as **one shared task lifecycle** feeding **three connected experiences**: a Student app, a Client app, and an Admin console. When one party acts, the others see it immediately — there are no divergent states.

The result is a working end-to-end product: a client funds a wallet, an admin assigns a verified student, the student delivers, the work is reviewed and approved, and payment is released — crediting the student and debiting the client automatically, with the task marked complete everywhere at once.

02 The problem

Talent and opportunity exist — but trust and payment don't connect them:

- **Students can't break in.** Skilled students have no track record, so clients won't risk hiring them — and they can't build a record without work.
- **Clients fear the risk.** Paying an unknown student upfront is risky; there's no guarantee the work lands or meets the brief.
- **There's no safe money rail.** Without escrow or oversight, neither side trusts the payment — so the deal never happens.

The core challenge was therefore not a feature — it was **making strangers comfortable enough to transact**.

03 The solution — four pillars

PILLAR	WHAT IT DELIVERS
Identity	Students earn a verified badge through an admin-reviewed check, so clients know exactly who they're hiring.
Discovery	Clients find the right student and students find real, relevant tasks — through bids, quick requests and rich search.
Escrow	Funds sit in escrow and are released only when work is approved — visible and legible, never a leap of faith.
Trust	Admin oversight, ratings and shared status make the whole exchange transparent to all three sides at once.

ROLE	WHAT THEY DO
Student	Gets verified, browses and bids on tasks, delivers work, gets paid, and builds a track record (projects, clients, rating, earnings).
Client	Funds a wallet, posts tasks, reviews and approves delivered work, and releases payment from escrow.
Admin	Verifies students, assigns/approves work, monitors escrow, resolves disputes, and rates the quality of each delivery.

One account can act as both a student and a client; the admin console is a separate, credential-gated experience.

Identity & verification

- Passwordless sign-up and login (email one-time code), with an embedded wallet created automatically on sign-up.
- A student/client toggle at login routes each user to the right experience.
- Student verification is an admin-reviewed request; students see a clear "verification in progress" state and are notified on approval. Verified status is what unlocks being assigned to work.

Student profile

- Real, live statistics: completed projects, distinct clients, average rating and review count, active bids, lifetime earnings, and verification status.
- Details entered at setup (university, department, reg number, skills, bio, LinkedIn, availability) are stored, displayed, and pre-filled in the edit form.

Discovery, search & bidding

- A "Find Work" marketplace of open tasks; each job card shows the client's **company name**.
- **Multi-field search** across title, description, job type, skills, company, budget and due date — matching every term entered.
- Students **place a bid** (own price, delivery time, pitch) or send a **quick request**. "My Bids" reflects their real bids.

The task lifecycle — synchronized across all three roles

- 1 **Client** posts a task → it awaits admin review.
 - 2 **Students** bid on or request the task.
 - 3 **Admin** assigns the best-fit student → escrow is marked funded.
 - 4 **Student** delivers the work (real file upload with previews).
 - 5 **Admin** approves the submission → hands it to the client.
 - 6 **Client** reviews and either requests changes (reopens the task, notifies the student) or approves.
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On approval, **payment releases** — student credited, client debited, task complete everywhere.

While a task still awaits admin review, the client can **edit or delete** it. After submission, the student can **view and edit their submission** rather than starting over.

Escrow & wallets

Money model: the client pays the task amount **plus a 5% platform fee**; the platform keeps the fee; the student receives the full task amount.

- Clients fund an in-app wallet; the platform wallet address and network are shown.
- On release, balances move atomically — client debited (amount + fee), student credited (amount), project and task completed, and a ledger entry recorded for both parties.
- Wallet balances, funded/spent totals and amounts on hold appear on the client dashboard and profile; students see balance and earnings.

Ratings, notifications & dashboards

- The client rates the delivered work (stars + written review); the admin can also rate quality. Ratings attach to the task and are shown to both the student and client.
- Per-user, **deep-linked** notifications open the exact screen where action is needed.
- Every dashboard has a **"Needs your attention"** section surfacing the next real action.

Admin console

- Credential-gated login that persists across refreshes for the session.
- A dashboard with a unified "Needs your attention" queue, a verification queue, task management & assignment, and escrow monitoring with a full payment breakdown, submission approval, release, refund and dispute handling, plus per-task ratings.
- **Live-activity detection:** background polling surfaces a non-blocking "New activity — Refresh" banner when a student or client changes something.
- **Sidebar count badges** for items needing attention (new tasks, pending verifications, submissions to review).

Marketing landing page

- A public landing page with a hero built from the real product dashboard, a statistics band, a photo showcase, testimonials, and tasteful animations.

LAYER	TECHNOLOGY
Frontend framework	React 18 + TypeScript
Build tool / dev server	Vite
Routing	React Router
Styling	CSS Modules; brand design tokens (blue, green, violet)
Typography	Manrope
Authentication	Privy — passwordless email one-time-code login + embedded wallets
Wallet network	Solana (embedded wallet addresses)
Backend & database	Supabase — PostgreSQL with the PostgREST auto REST API
Access control	PostgreSQL Row-Level Security
File storage	Supabase Storage (student work submissions)
Server-side logic	PostgreSQL functions / RPC (wallet funding & atomic payment release)
Hosting & delivery	Netlify (static SPA hosting, CLI deployments)
Version control	Git / GitHub
Design	Figma (wireframes, screen flows, product showcase)

07 System architecture

A single-page React application talking directly to Supabase.

- The **frontend** (React + Vite) is a static bundle served by Netlify, with client-side routing and a redirect rule so every route resolves to the app.
- **Supabase** provides the database, an auto-generated REST API, file storage and server-side functions, reached through a typed data-access layer so query shapes live in one place.
- **Authentication** is handled by Privy; each user maps to one row in the *profiles* table.
- **Money movement** runs through two PostgreSQL functions — fund a wallet, and release payment atomically (debit client, credit student, complete task, write ledger) in one transaction.
- **File submissions** are uploaded to a Supabase Storage bucket and referenced by URL for client and admin preview.

08 Data model

TABLE	PURPOSE
profiles	One row per user — email, role, wallet address, simulated wallet balance.
student_profiles	Student details — university, department, skills, portfolio, verification.
client_profiles	Client details — company name, about, verification.
tasks	Posted work — title, category, budget, skills, due date, status.
task_attachments	Files attached to a task brief.
bids	A student's bid or request on a task.
projects	A task assigned to a student — amount, status, payment status, work rating.
submissions	Work delivered on a project (files + note).
notifications	Per-user notifications with deep links.
wallet_transactions	The simulated wallet ledger (funding, payments, earnings).

Schema changes are managed as ordered SQL migration files (initial schema, access policies, wallets & payment functions, notification links, per-task ratings).

09 Security, deployment & process

Security & access

- Passwordless authentication (email one-time code) — no stored passwords.
- Row-Level Security enabled on every table; production intent is to enforce per-user ownership through the identity bridge.
- The admin console is credential-gated, required once per browser session.
- Secrets are supplied via environment variables, never committed.

Deployment & delivery

- Hosted on Netlify as a static SPA and **live** at skillbridge-africa-193.netlify.app.
- Deployed by building the app and publishing through the Netlify CLI; environment config injected at build time.

Design process

- Research & framing → wireframes → high-fidelity screens & flows → a reusable design system → a product case study & showcase, all produced in Figma.

Team & collaboration

- Tracked in a shared Git/GitHub repository with multiple contributors, feature branches and pull requests, and a clear, descriptive commit history.

A working product — not just screens. Student, client and admin experiences are live and function end-to-end, including the complete escrow loop (fund → assign → deliver → approve → release), a real admin console, real file uploads, and attention-first dashboards.

Roadmap

- Move escrow **on-chain** for real, auditable settlement.
- Feed ratings into **discovery and ranking**.
- Add **in-app messaging** between clients and students.
- Add a **dispute-resolution** workflow.
- Harden **row-level security** for production launch.

SkillBridge Africa — connecting African university talent with the clients who need it, safely.